

Lab-4 PHP

Q1-Create a PHP file with a namespace Library containing a class Book with properties title and author. The class should have a method getDetails() to return book details. Then, in another file, use this class.

Code:

Lab4.php

```
<?php
namespace Library;

class Book {
    public string $title;
    public string $author;

    public function __construct(string $title, string $author) {
        $this->title = $title;
        $this->author = $author;
    }

    public function getDetails(): string {
        return "Title: {$this->title}, Author: {$this->author}";
    }
}
```

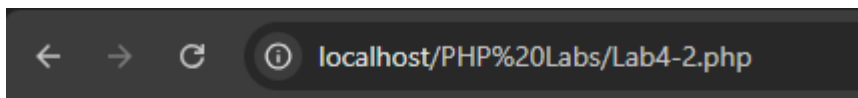
Lab4-2.php

```
<?php
require_once 'Lab4.php';

use Library\Book;

$book = new Book('The Alchemist', 'Paulo Coelho');
echo $book->getDetails();
```

Output:



Title: The Alchemist, Author: Paulo Coelho

Q2-Write a PHP program that takes two numbers and an operator (+, -, *, /) from the user and performs the corresponding operation using a switch statement.

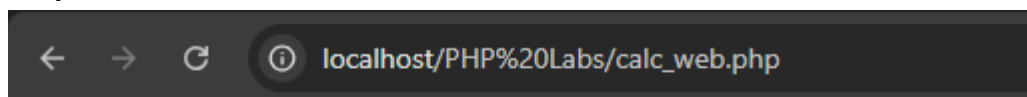
Code:

```
<!-- calc_form.html -->
<form method="post" action="calc_web.php">
  <input name="a" type="number" step="any" required placeholder="First number">
  <select name="op">
    <option>+</option><option>-</option><option>*</option><option>/</option>
  </select>
  <input name="b" type="number" step="any" required placeholder="Second number">
  <button type="submit">Compute</button>
</form>
```

```
<?php
// calc_web.php
if ($_SERVER['REQUEST_METHOD'] === 'POST') {
    $a = (float) $_POST['a'];
    $op = $_POST['op'];
    $b = (float) $_POST['b'];

    switch ($op) {
        case '+': $res = $a + $b; break;
        case '-': $res = $a - $b; break;
        case '*': $res = $a * $b; break;
        case '/':
            if ($b == 0.0) { echo "Division by zero error."; exit; }
            $res = $a / $b; break;
        default: echo "Invalid operator"; exit;
    }
    echo "Result: $res";
}
```

Output:



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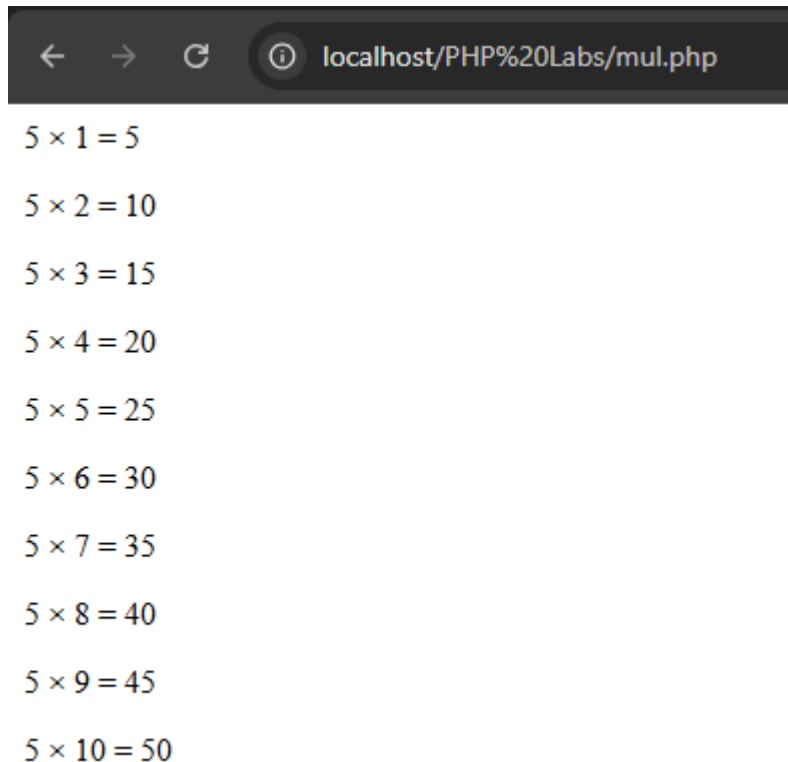
Result: 10

Q3-Write a PHP script that asks the user for a number and prints its multiplication table (up to 10) using a for loop.

Code:

```
<?php
$n = intval($_GET['n'] ?? 5);
for ($i=1;$i<=10;$i++){
    echo "<p>$n × $i = " . ($n*$i) . "</p>";
}
```

Output:



Q4-Write a PHP program that calculates the sum of the first N natural numbers using a for loop.

Code:

```
<!DOCTYPE html>
<html>
<head>
    <title>Sum of N Natural Numbers</title>
</head>
<body>

<form method="post">
    Enter N (natural number):
    <input type="number" name="n" min="1" required>
```

```

        <button type="submit">Calculate</button>
    </form>

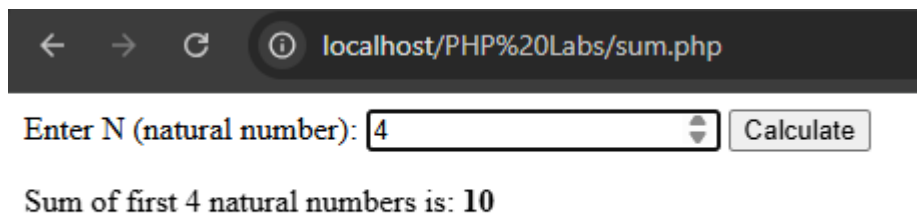
    <?php
    if (isset($_POST['n'])) {
        $n = (int) $_POST['n'];

        if ($n < 1) {
            echo "<p>Enter a positive integer.</p>";
        } else {
            $sum = 0;
            for ($i = 1; $i <= $n; $i++) {
                $sum += $i;
            }
            echo "<p>Sum of first $n natural numbers is: <b>$sum</b></p>";
        }
    }
    ?>

</body>
</html>

```

Output:



Enter N (natural number):

Sum of first 4 natural numbers is: 10

Q5-Write a PHP script to reverse a given number using a while loop.

Code:

```

<!DOCTYPE html>
<html>
<head>
    <title>Reverse Number</title>
</head>
<body>

    <form method="post">
        Enter an integer to reverse:
        <input type="number" name="num" required>
        <button type="submit">Reverse</button>
    </form>

```

```

<?php
if (isset($_POST['num'])) {
    $input = $_POST['num'];

    if (!is_numeric($input)) {
        echo "<p>Please enter a number.</p>";
    } else {
        $negative = false;
        if ($input < 0) {
            $negative = true;
            $input = abs($input);
        }

        $num = (int)$input;
        $rev = 0;

        while ($num > 0) {
            $rev = $rev * 10 + ($num % 10);
            $num = intval($num / 10);
        }

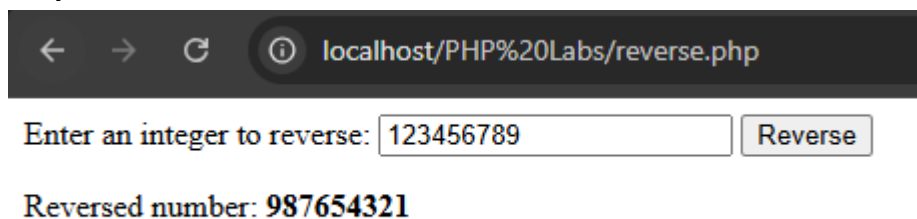
        if ($negative) $rev = -$rev;

        echo "<p>Reversed number: <b>$rev</b></p>";
    }
}
?>

```

</body>
</html>

Output:



Enter an integer to reverse:

Reversed number: **987654321**